

1

M

Write a program that displays the Netsal(Sal+comm) of an employee entered at run time from EMP table.(Use NVL Function).

Program with
Database
Server



```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     v_empno  employees.employee_id%TYPE := &employee_id;
  3     v_ename  employees.first_name%TYPE;
  4     v_sal    employees.salary%TYPE;
  5     v_comm   employees.commission_pct%TYPE;
  6     v_netsal NUMBER;
  7 BEGIN
  8     SELECT first_name, salary, NVL(commission_pct,0)
  9     INTO v_ename, v_sal, v_comm
 10     FROM employees
 11     WHERE employee_id = v_empno;
 12
 13     v_netsal := v_sal + v_comm;
 14
 15     DBMS_OUTPUT.PUT_LINE('Employee Number : ' || v_empno);
 16     DBMS_OUTPUT.PUT_LINE('Employee Name   : ' || v_ename);
 17     DBMS_OUTPUT.PUT_LINE('Salary        : ' || v_sal);
 18     DBMS_OUTPUT.PUT_LINE('Commission   : ' || v_comm);
 19     DBMS_OUTPUT.PUT_LINE('Net Salary    : ' || v_netsal);
 20
 21 EXCEPTION
 22     WHEN NO_DATA_FOUND THEN
 23         DBMS_OUTPUT.PUT_LINE('Employee does not exist.');
```

Enter value for employee_id: 100

```
old 2:     v_empno  employees.employee_id%TYPE := &employee_id;
new 2:     v_empno  employees.employee_id%TYPE := 100;
Employee Number : 100
Employee Name   : Steven
Salary         : 24000
Commission     : 0
Net Salary      : 24000

PL/SQL procedure successfully completed.

SQL>
```

1

M

Write a program that will display the entire row by passing the empno. (Use %ROWTYPE)

Working with
Composite
Data types



```

SQL> set serveroutput on;
SQL> declare
2     v_empno employees.employee_id%TYPE := &employee_id;
3     emp_rec employees%rowtype;
4 begin
5     select *
6     into emp_rec
7     from employees
8     where employee_id=v_empno;
9     DBMS_OUTPUT.PUT_LINE('Employee Number : ' || emp_rec.employee_id);
10    DBMS_OUTPUT.PUT_LINE('Employee Name : ' || emp_rec.first_name);
11    DBMS_OUTPUT.PUT_LINE('Job : ' || emp_rec.job_id);
12    DBMS_OUTPUT.PUT_LINE('Manager : ' || emp_rec.manager_id);
13    DBMS_OUTPUT.PUT_LINE('Hire Date : ' || emp_rec.hire_date);
14    DBMS_OUTPUT.PUT_LINE('Salary : ' || emp_rec.salary);
15    DBMS_OUTPUT.PUT_LINE('Commision : ' || nvl(emp_rec.commission_pct,0));
16    DBMS_OUTPUT.PUT_LINE('Department No : ' || emp_rec.department_id);
17 Exception
18     when NO_DATA_FOUND then
19         dbms_output.put_line('Employee Not Found. ');
20 end;
21 /
Enter value for employee_id: 100
old 2:     v_empno employees.employee_id%TYPE := &employee_id;
new 2:     v_empno employees.employee_id%TYPE := 100;
Employee Number : 100
Employee Name : Steven
Job : AD_PRES
Manager :
Hire Date : 17-JUN-03
Salary : 24000
Commision : 0
Department No : 90

PL/SQL procedure successfully completed.

```

No.	Cat	Hands-on Assignment	Topics Covered	Status
1	M	Write a implicit cursor that will prompt for a JOB and delete the employees working in that Job. Store the number of records deleted in a session variable.	Using Explicit Cursors	☐
2	M	Rollback the the previous Delete Statement.	Using Explicit Cursors	☐
3	S	Write a program that displays the Top N salaries using simple Explicit Cursors.	Using Explicit Cursors	☐
4	M	Create a Simple Explicit Cursor that displays all the employees who are earning more than the average salary of the Emp Table.	Using Explicit Cursors	☐

5	M	Create a Simple Explicit cursor that displays the employees who earn more than 2000 and have joined the company after 15-Jun-1981. Print the output : <> earns <> and joined the organization on <>.	Using Explicit Cursors	☐
6	S	Create a Simple Explicit cursor that displays the hiredate in the format DD-MM-RRRR and Day. Sort the records on DAY starting from Saturday.	Using Explicit Cursors	☐
7	S	Create a simple explicit cursor that displays the employees names and commission amounts. If an employee does not earn commission, show "No Commission" . Label the column COMM.	Using Explicit Cursors	☐
8	M	Create a parameter cursor that takes DEPTNO as a parameter and update the star column with '*'. Every '*' represent a 1000. For example : If the employees salary is 5000 then update star column with '*****'. Use Parameter Cursor ,CURSOR WITH FOR LOOP and WHERE CURRENT OF OPTION.	Using Explicit Cursors	☐
9	M	Write a Parameter Cursor program that promotes CLERK who earn more than 1000 to SR CLERK and increase the salary by 10%. Pass CLERK as a parameter to the Cursor. Use Parameter Cursor, CUSOR WITH FOR LOOP and CURSOR WITH UPDATE CLAUSE.	Using Explicit Cursors	☐
10	S	Display entire record from the Dept table by passing a DEPTNO. Increase the Salary of employees working in deptno 10 by 15%, Deptno 20 by 15% and others by 5% Also display the corresponding the employees working in that Dept. Use a parameter Cursor , For Loop Cursor and Cursor with Update clause.	Using Explicit Cursors	☐
11	M	Write a program that will prompt the user for a CHOICE. IF CHOICE is 1 then display all the employees who earn more than 2000 from EMP table else display all the Employees who earn less than 2000. Use REF CURSORS.	Using Explicit Cursors	☐

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2     V_JOB_ID EMPLOYEES.JOB_ID%TYPE := UPPER('&JOB_ID');
3     V_COUNT  NUMBER;
4 BEGIN
5     DELETE FROM EMPLOYEES
6     WHERE JOB_ID = V_JOB_ID;
7
8     V_COUNT := SQL%ROWCOUNT;
9
10    DBMS_OUTPUT.PUT_LINE(V_COUNT || ' Record(s) Deleted.');
```

11

```

12    COMMIT;
13 END;
14 /
Enter value for job_id: SH_CLERK
old 2:     V_JOB_ID EMPLOYEES.JOB_ID%TYPE := UPPER('&JOB_ID');
new 2:     V_JOB_ID EMPLOYEES.JOB_ID%TYPE := UPPER('SH_CLERK');
20 Record(s) Deleted.

PL/SQL procedure successfully completed.
```

2. Rollback the Previous DELETE Statement

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> BEGIN
2     ROLLBACK;
3     DBMS_OUTPUT.PUT_LINE('Rollback Successful');
4 END;
5 /
Rollback Successful

PL/SQL procedure successfully completed.

SQL> |
```

3. Display Top N Salaries Using Explicit Cursor

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     V_N NUMBER := &N;
  3     V_COUNT NUMBER := 0;
  4
  5     CURSOR C1 IS
  6         SELECT EMPLOYEE_ID, FIRST_NAME, SALARY
  7         FROM EMPLOYEES
  8         ORDER BY SALARY DESC;
  9
 10 BEGIN
 11     FOR REC IN C1 LOOP
 12         EXIT WHEN V_COUNT = V_N;
 13
 14         DBMS_OUTPUT.PUT_LINE(
 15             REC.EMPLOYEE_ID || ' ' ||
 16             REC.FIRST_NAME || ' ' ||
 17             REC.SALARY);
 18
 19         V_COUNT := V_COUNT + 1;
 20     END LOOP;
 21 END;
 22 /
Enter value for n: 5
old 2:      V_N NUMBER := &N;
new 2:      V_N NUMBER := 5;
100 Steven 24000
101 Neena 17000
102 Lex 17000
145 John 14000
146 Karen 13500

PL/SQL procedure successfully completed.

```

4.

Display Employees Earning More Than Average Salary (Explicit Cursor)

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2   V_AVG EMPLOYEES.SALARY%TYPE;
  3
  4   V_EMPNO EMPLOYEES.EMPLOYEE_ID%TYPE;
  5   V_ENAME EMPLOYEES.FIRST_NAME%TYPE;
  6   V_SAL EMPLOYEES.SALARY%TYPE;
  7
  8   CURSOR C1 IS
  9       SELECT EMPLOYEE_ID, FIRST_NAME, SALARY
10       FROM EMPLOYEES
11       WHERE SALARY > V_AVG;
12
13 BEGIN
14     SELECT AVG(SALARY)
15     INTO V_AVG
16     FROM EMPLOYEES;
17
18     DBMS_OUTPUT.PUT_LINE('Average Salary = ' || ROUND(V_AVG,2));
19
20     OPEN C1;
21
22     LOOP
23         FETCH C1 INTO
24             V_EMPNO,
25             V_ENAME,
26             V_SAL;
27
28         EXIT WHEN C1%NOTFOUND;
29
30         DBMS_OUTPUT.PUT_LINE(
31             V_EMPNO || ' ' ||
32             V_ENAME || ' ' ||
33             V_SAL);
34     END LOOP;
35
36     CLOSE C1;
37 END;
38 /
Average Salary = 7208.23
100 Steven 24000
101 Neena 17000
102 Lex 17000
103 Alexander 9000
108 Nancy 12008
109 Daniel 9000
110 John 8200
111 Ismael 7700
112 Jose Manuel 7800
114 Den 11000
120 Matthew 8000
121 Adam 8200
122 Payam 7900
145 John 14000
146 Karen 13500
147 Alberto 12000
148 Gerald 11000
149 Eleni 10500
150 Peter 10000
151 David 9500
152 Peter 9000
153 Christopher 8000
154 Nanette 7500

```

5. Display Employees Who Earn More Than 2000 and Joined After 15-JUN-1981 Using Simple Explicit Cursor

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2   CURSOR C1 IS
3     SELECT FIRST_NAME, SALARY, HIRE_DATE
4     FROM EMPLOYEES
5     WHERE SALARY > 2000
6     AND HIRE_DATE > TO_DATE('15-JUN-2000','DD-MM-YYYY');
7
8   V_NAME      EMPLOYEES.FIRST_NAME%TYPE;
9   V_SAL       EMPLOYEES.SALARY%TYPE;
10  V_HIREDATE   EMPLOYEES.HIRE_DATE%TYPE;
11
12 BEGIN
13   OPEN C1;
14
15   LOOP
16     FETCH C1 INTO V_NAME, V_SAL, V_HIREDATE;
17     EXIT WHEN C1%NOTFOUND;
18
19     DBMS_OUTPUT.PUT_LINE(
20       V_NAME || ' ' || V_SAL || ' ' ||
21       ' ' || V_HIREDATE || ' ' ||
22       TO_CHAR(V_HIREDATE, 'DD-MM-YYYY')
23     );
24   END LOOP;
25
26   CLOSE C1;
27 END;
28 /
Steven earns 24000 and joined the organization on 17-JUN-2003
Neena earns 17000 and joined the organization on 21-SEP-2005
Alexander earns 9000 and joined the organization on 03-JAN-2006
Duke earns 8000 and joined the organization on 21-MAY-2007
David earns 8000 and joined the organization on 25-JAN-2005
Valli earns 8000 and joined the organization on 05-FEB-2006
Diana earns 9000 and joined the organization on 07-FEB-2007
John earns 8200 and joined the organization on 28-SEP-2005
Ismael earns 7700 and joined the organization on 26-SEP-2005
Jose Manuel earns 7600 and joined the organization on 07-MAR-2006
Luis earns 6000 and joined the organization on 07-OCT-2007
Shelli earns 2000 and joined the organization on 24-OCT-2005
Sigal earns 2000 and joined the organization on 24-JUL-2005
Guy earns 2600 and joined the organization on 15-NOV-2006
Karen earns 2500 and joined the organization on 10-DEC-2007
Matthew earns 2000 and joined the organization on 18-JUL-2004
Adam earns 8200 and joined the organization on 18-APR-2005
Shanta earns 6500 and joined the organization on 19-OCT-2005
Kevin earns 5500 and joined the organization on 16-NOV-2007
Julia earns 2200 and joined the organization on 16-JUL-2005
Irene earns 2700 and joined the organization on 25-OCT-2006
James earns 2400 and joined the organization on 14-JAN-2007
Steven earns 2200 and joined the organization on 06-MAR-2006
Laura earns 2200 and joined the organization on 20-AUG-2005
Neha earns 2000 and joined the organization on 29-OCT-2005
Lexa earns 2200 and joined the organization on 16-FEB-2005
TJ earns 2100 and joined the organization on 18-APR-2007
Jason earns 2200 and joined the organization on 14-JUL-2004
Michael earns 2000 and joined the organization on 26-AUG-2006
MI earns 2400 and joined the organization on 12-OCT-2007
Hazel earns 2200 and joined the organization on 06-FEB-2006
Doris earns 2000 and joined the organization on 10-JUL-2005
Stephen earns 2200 and joined the organization on 26-OCT-2005
John earns 2000 and joined the organization on 12-FEB-2006
Joshua earns 2200 and joined the organization on 06-APR-2006
Trenna earns 2000 and joined the organization on 17-OCT-2003
Curtis earns 2100 and joined the organization on 20-JAN-2005
Terry earns 2000 and joined the organization on 15-MAR-2006
Peter earns 2000 and joined the organization on 09-JUL-2006
John earns 18000 and joined the organization on 01-OCT-2004
Maren earns 12000 and joined the organization on 05-JAN-2005
Alberto earns 12000 and joined the organization on 18-MAR-2005
Gerald earns 11000 and joined the organization on 15-OCT-2007
Clara earns 10000 and joined the organization on 23-JAN-2006
Peter earns 10000 and joined the organization on 28-JAN-2005
David earns 9500 and joined the organization on 24-MAR-2005
Peter earns 9000 and joined the organization on 29-AUG-2005
Christopher earns 8500 and joined the organization on 20-MAR-2006
Doris earns 7500 and joined the organization on 09-APR-2006
Oliver earns 7000 and joined the organization on 23-NOV-2007
Javette earns 10000 and joined the organization on 10-JAN-2004
Patrick earns 9500 and joined the organization on 04-MAR-2004
Allan earns 9000 and joined the organization on 01-AUG-2004
Lindsey earns 8000 and joined the organization on 19-MAR-2005
Louise earns 7000 and joined the organization on 15-OCT-2005
Sarath earns 7000 and joined the organization on 02-NOV-2005
Clara earns 10000 and joined the organization on 11-NOV-2005
Danielle earns 9500 and joined the organization on 19-MAR-2007
Mettea earns 7200 and joined the organization on 24-JAN-2006
David earns 6500 and joined the organization on 23-FEB-2006
Gardner earns 6400 and joined the organization on 24-MAR-2006
Elli earns 6200 and joined the organization on 21-APR-2006
Lisa earns 11500 and joined the organization on 11-MAR-2005
Harrison earns 10000 and joined the organization on 23-MAR-2006
Tayler earns 9000 and joined the organization on 24-JAN-2006
William earns 7000 and joined the organization on 23-FEB-2007
Elizabeth earns 7000 and joined the organization on 24-MAR-2007
Gudita earns 6100 and joined the organization on 21-APR-2006
Ellen earns 11000 and joined the organization on 11-MAY-2004
Alyssa earns 8000 and joined the organization on 19-MAR-2005
Jonathan earns 5000 and joined the organization on 24-MAR-2006
Jack earns 8400 and joined the organization on 23-APR-2006
Kristen earns 7000 and joined the organization on 24-MAY-2007
Charles earns 6200 and joined the organization on 04-JAN-2006
Jennifer earns 10000 and joined the organization on 17-SEP-2003
Michael earns 12000 and joined the organization on 17-FEB-2004
Pat earns 6000 and joined the organization on 17-AUG-2005

PL/SQL procedure successfully completed.

SQL> |

```

6. Display Hire Date in DD-MM-RRRR and Day Format (Sorted from Saturday)

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2     CURSOR C1 IS
3         SELECT FIRST_NAME,
4                TO_CHAR(HIRE_DATE, 'DD-MM-RRRR') HDATE,
5                TO_CHAR(HIRE_DATE, 'DAY') DAYNAME
6         FROM EMPLOYEES
7         ORDER BY TO_CHAR(HIRE_DATE, 'D');
8
9     V_ENAME EMPLOYEES.FIRST_NAME%TYPE;
10    V_HDATE VARCHAR2(20);
11    V_DAY   VARCHAR2(20);
12
13 BEGIN
14     OPEN C1;
15
16     LOOP
17         FETCH C1 INTO V_ENAME, V_HDATE, V_DAY;
18         EXIT WHEN C1%NOTFOUND;
19
20         DBMS_OUTPUT.PUT_LINE(
21             RPAD(V_ENAME,15) || ' ' ||
22             V_HDATE || ' ' ||
23             RTRIM(V_DAY)
24         );
25     END LOOP;
26
27     CLOSE C1;
28 END;
29 /
John      12-02-2006 SUNDAY
Jack      23-04-2006 SUNDAY
Peter     09-07-2006 SUNDAY
Valli     05-02-2006 SUNDAY
Peter     30-01-2005 SUNDAY
Mozhe     30-10-2005 SUNDAY
Sigal     24-07-2005 SUNDAY
Matthew   18-07-2004 SUNDAY

```

7. Display Employee Names and Commission (Show "No Commission")


```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     CURSOR C1 IS
  3         SELECT FIRST_NAME,
  4                NVL(TO_CHAR(COMMISSION_PCT), 'No Commission') COMM
  5         FROM EMPLOYEES;
  6
  7     V_ENAME EMPLOYEES.FIRST_NAME%TYPE;
  8     V_COMM  VARCHAR2(30);
  9
 10 BEGIN
 11     OPEN C1;
 12
 13     LOOP
 14         FETCH C1 INTO V_ENAME, V_COMM;
 15         EXIT WHEN C1%NOTFOUND;
 16
 17         DBMS_OUTPUT.PUT_LINE(
 18             RPAD(V_ENAME,15) || ' COMM : ' || V_COMM
 19         );
 20     END LOOP;
 21
 22     CLOSE C1;
 23 END;
 24 /
Steven          COMM : No Commission
Neena           COMM : No Commission
Lex             COMM : No Commission
Alexander       COMM : No Commission
Bruce          COMM : No Commission
David           COMM : No Commission
Valli          COMM : No Commission
Diana          COMM : No Commission
Nancy          COMM : No Commission
Daniel         COMM : No Commission

```

8. Parameter Cursor to Update STAR Column

Requirement

- Pass **DEPTNO** as a parameter.
- Every * represents **1000** salary.
- Salary = 5000 → *****
- Use **Parameter Cursor**, **Cursor FOR LOOP**, **WHERE CURRENT OF**.

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     V_DEPTNO EMPLOYEES.DEPARTMENT_ID%TYPE := &DEPARTMENT_ID;
  3
  4     CURSOR C1(P_DNO EMPLOYEES.DEPARTMENT_ID%TYPE) IS
  5         SELECT SALARY, STAR
  6         FROM EMPLOYEES
  7         WHERE DEPARTMENT_ID = P_DNO
  8         FOR UPDATE OF STAR;
  9
 10 BEGIN
 11     FOR REC IN C1(V_DEPTNO) LOOP
 12
 13         UPDATE EMPLOYEES
 14         SET STAR = RPAD('*', TRUNC(REC.SALARY/1000), '*')
 15         WHERE CURRENT OF C1;
 16
 17     END LOOP;
 18
 19     COMMIT;
 20
 21     DBMS_OUTPUT.PUT_LINE('STAR column updated successfully.');
```

Enter value for department_id: 10

```

old 2:     V_DEPTNO EMPLOYEES.DEPARTMENT_ID%TYPE := &DEPARTMENT_ID;
new 2:     V_DEPTNO EMPLOYEES.DEPARTMENT_ID%TYPE := 10;
STAR column updated successfully.

PL/SQL procedure successfully completed.
```

9. Promote CLERK to SR CLERK

Requirement

- Pass **JOB** as parameter.
- Promote **CLERK** earning more than **1000**
- Increase salary by **10%**
- Use **Parameter Cursor**
- Use **Cursor FOR LOOP**
- Use **FOR UPDATE** and **WHERE CURRENT OF**

Program

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2      V_JOB EMPLOYEES.JOB_ID%TYPE := '&JOB_ID';
  3
  4      CURSOR C1(P_JOB EMPLOYEES.JOB_ID%TYPE) IS
  5          SELECT JOB_ID, SALARY
  6          FROM EMPLOYEES
  7          WHERE JOB_ID = P_JOB
  8              AND SALARY > 1000
  9          FOR UPDATE OF SALARY;
10
11 BEGIN
12     FOR REC IN C1(V_JOB) LOOP
13
14         UPDATE EMPLOYEES
15         SET SALARY = SALARY * 1.10
16         WHERE CURRENT OF C1;
17
18     END LOOP;
19
20     COMMIT;
21
22     DBMS_OUTPUT.PUT_LINE('Salary Updated Successfully.');
```

23

```

24 END;
25 /
Enter value for job_id: SA_REP
old 2:      V_JOB EMPLOYEES.JOB_ID%TYPE := '&JOB_ID';
new 2:      V_JOB EMPLOYEES.JOB_ID%TYPE := 'SA_REP';
Salary Updated Successfully.

PL/SQL procedure successfully completed.
```

10. Display Department Record and Update Employee Salaries

Requirement

- Pass DEPTNO
- Display DEPT table record
- Increase salary
 - Dept 10 → 15%
 - Dept 20 → 15%
 - Others → 5%
- Display corresponding employees
- Use Parameter Cursor
- Cursor FOR LOOP

- Cursor WITH UPDATE clause

Program

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2   V_DEPTNO EMPLOYEES.DEPARTMENT_ID%TYPE := &DEPARTMENT_ID;
3
4   V_DNAME DEPARTMENTS.DEPARTMENT_NAME%TYPE;
5   V_LOCID DEPARTMENTS.LOCATION_ID%TYPE;
6
7   CURSOR C1(P_DNO EMPLOYEES.DEPARTMENT_ID%TYPE) IS
8       SELECT EMPLOYEE_ID,
9              FIRST_NAME,
10             SALARY,
11             DEPARTMENT_ID
12       FROM EMPLOYEES
13       WHERE DEPARTMENT_ID = P_DNO
14       FOR UPDATE OF SALARY;
15
16 BEGIN
17     SELECT DEPARTMENT_NAME, LOCATION_ID
18     INTO V_DNAME, V_LOCID
19     FROM DEPARTMENTS
20     WHERE DEPARTMENT_ID = V_DEPTNO;
21
22     DBMS_OUTPUT.PUT_LINE('Department ID   : ' || V_DEPTNO);
23     DBMS_OUTPUT.PUT_LINE('Department Name : ' || V_DNAME);
24     DBMS_OUTPUT.PUT_LINE('Location ID    : ' || V_LOCID);
25     DBMS_OUTPUT.PUT_LINE('-----');
26
27     FOR REC IN C1(V_DEPTNO) LOOP
28
29         IF REC.DEPARTMENT_ID = 10 THEN
30
31             UPDATE EMPLOYEES
32             SET SALARY = SALARY * 1.15
33             WHERE CURRENT OF C1;
34
35             ELSIF REC.DEPARTMENT_ID = 20 THEN
36
37                 UPDATE EMPLOYEES
38                 SET SALARY = SALARY * 1.15
39                 WHERE CURRENT OF C1;
40
41             ELSE
42
43                 UPDATE EMPLOYEES
44                 SET SALARY = SALARY * 1.05
45                 WHERE CURRENT OF C1;
46
47             END IF;
48
49             DBMS_OUTPUT.PUT_LINE(
50                 REC.EMPLOYEE_ID || ' ' ||
51                 REC.FIRST_NAME || ' Salary Updated'
52             );
53
54         END LOOP;
55
56     COMMIT;
57
58 END;
59 /
Enter value for department_id: 10
old 2:   V_DEPTNO EMPLOYEES.DEPARTMENT_ID%TYPE := &DEPARTMENT_ID;
new 2:   V_DEPTNO EMPLOYEES.DEPARTMENT_ID%TYPE := 10;
Department ID   : 10
Department Name : Administration
Location ID     : 1700
-----
200 Jennifer Salary Updated

PL/SQL procedure successfully completed.

```

11: Use REF CURSOR

Requirement

- Prompt the user for a **CHOICE**.
- If **CHOICE = 1**, display employees whose **SAL > 2000**.
- Otherwise, display employees whose **SAL < 2000**.

- Use REF CURSOR.

Program

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     CHOICE NUMBER := &CHOICE;
  3
  4     TYPE EMP_CUR IS REF CURSOR;
  5     C1 EMP_CUR;
  6
  7     V_EMPNO EMPLOYEES.EMPLOYEE_ID%TYPE;
  8     V_ENAME EMPLOYEES.FIRST_NAME%TYPE;
  9     V_SAL   EMPLOYEES.SALARY%TYPE;
10
11 BEGIN
12
13     IF CHOICE = 1 THEN
14         OPEN C1 FOR
15             SELECT EMPLOYEE_ID, FIRST_NAME, SALARY
16             FROM EMPLOYEES
17             WHERE SALARY > 2000;
18     ELSE
19         OPEN C1 FOR
20             SELECT EMPLOYEE_ID, FIRST_NAME, SALARY
21             FROM EMPLOYEES
22             WHERE SALARY < 2000;
23     END IF;
24
25     LOOP
26         FETCH C1 INTO V_EMPNO, V_ENAME, V_SAL;
27         EXIT WHEN C1%NOTFOUND;
28
29         DBMS_OUTPUT.PUT_LINE(
30             'Employee ID : ' || V_EMPNO ||
31             ' Name : ' || V_ENAME ||
32             ' Salary : ' || V_SAL
33         );
34     END LOOP;
35
36     CLOSE C1;
37
38 END;
39 /
Enter value for choice: 1
old 2:     CHOICE NUMBER := &CHOICE;
new 2:     CHOICE NUMBER := 1;
Employee ID : 100 Name : Steven Salary : 24000
Employee ID : 101 Name : Neena Salary : 17000
Employee ID : 102 Name : Lex Salary : 17000
Employee ID : 103 Name : Alexander Salary : 9000
Employee ID : 104 Name : Bruce Salary : 6000
Employee ID : 105 Name : David Salary : 4800
Employee ID : 106 Name : Valli Salary : 4800
Employee ID : 107 Name : Diana Salary : 4200
Employee ID : 108 Name : Nancy Salary : 12008
Employee ID : 109 Name : Daniel Salary : 9000
Employee ID : 110 Name : John Salary : 8200
Employee ID : 111 Name : Ismael Salary : 7700
Employee ID : 112 Name : Jose Manuel Salary : 7800
Employee ID : 113 Name : Luis Salary : 6900
Employee ID : 114 Name : Den Salary : 11000

```